Surendra Kumar Chandrasekaran

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Education

University of Maryland – College Park *Sep 2025 – May 2027*
M.S. in Applied Machine Learning **GPA: 4.0/4.0**
Relevant Coursework: Principles of Machine Learning, Data Science, Probability and Statistics

College of Engineering, Guindy (Anna University) *Nov 2021 – May 2025*
B.E. in Computer Science and Engineering **GPA: 3.66/4**
Relevant Coursework: Data Structures and Algorithms, Computer Architecture, Operating Systems, DBMS

Skills

Programming: Python, C/C++, Java, SQL, JAX
Machine Learning: Deep Learning, NLP, RAG, Transformers, Computer Vision, Model Optimization, MLOps
Frameworks & Libraries: PyTorch, TensorFlow, HuggingFace, scikit-learn, CUDA, NumPy, Pandas, LangChain
Tools & Databases: Git, Docker, AWS, GCP, Linux, FastAPI, ReactJS, PySpark, MySQL, MongoDB, FAISS

Experience

Graduate Research Assistant – University of Maryland *Sep 2025 – Present*
- Built an NLP pipeline for political discourse analysis using a **retrieval-augmented architecture**, achieving **98% recall** and **100% precision** with **FAISS** and **GPT-4**.
- Developed scalable evaluation frameworks for multimodal content analysis, combining text and computer vision techniques to detect coded language patterns and enable cross-modal signal detection.

Founding Engineer – EyeZenX Analytics Pvt. Ltd. *Feb 2025 – Aug 2025*
- Spearheaded the design and deployment of a full-stack ML fundus image analysis system using **AWS EC2** (for models), **Flask**, and **ReactJS** on **Render**, boosting cross-platform interoperability and raising diagnostic success rates by **25%**
- Optimized deep learning pipelines for fundus image analysis using **PyTorch** for **EfficientNet**, **MobileNetV3**, and **U-Net** models for disease classification, image quality assessment, and microvascular segmentation from fundus images

Machine Learning Intern – Kaar Technologies *May 2024 – Feb 2025*
- Implemented Multi-Document QA RAG pipelines using **LangChain, FAISS,** and **TinyLLaMA-1.1B**, reducing query response latency by **35%** while maintaining efficient inference using prompt engineering
- Automated resume-driven interviews by developing a scalable GenAI Recruitment Portal with **FastAPI** and **ReactJS**, integrating **Gemini Pro** for LLM Inferencing to accelerate candidate evaluation and streamline recruitment workflows.
- Devised a lightweight **KPI retrieval** system leveraging classical NLP techniques with **Word2Vec** and **spaCy**, enabling efficient extraction and classification of key performance indicators while cutting processing time by **40%**.

Projects

AutoCRM: Explainable Agentic CRM Automation *Jan 2025 – Apr 2025*
- Orchestrated a **multi-agent B2C customer pipeline** using **CrewAI** and **GPT-4o-mini** with agents for customer scoring, personalized outreach, critic-based self-correction, and churn escalation, each outputting justifications for every decision.
- Built a **business-configurable reasoning layer** via **Streamlit** letting users adjust customer prioritization, scoring weights, and outreach tone, dynamically rewriting agent system prompts.

CaseSage: Analysis of Indian Legal Documents *Nov 2024 – May 2025*
- Built a Legal NLP system for Indian legal judgments implementing hybrid summarization using **Longformer** for abstractive and **NER-based Entity Ranking** for extractive summarization, improving **ROUGE Score by 50%** over benchmark.
- Delivered precedent retrieval with **Knowledge Graphs** via **Neo4j** (**+40%** accuracy), RAG-based QA with **DeepSeek-v3** (**-65%** lookup time), and automated semantic segmentation of legal terms with **LSTMs**.

GraphMind: Agentic TensorFlow Debugger *Sep 2025 - Oct 2025*
- Implemented an AI-driven **TensorFlow** debugging system using **Graph-based RAG**, **FAISS**, **LangGraph**, and **NVIDIA Nemotron**, streamlining error analysis and enhancing code reasoning efficiency.
- Introduced self-learning feedback loops and agent-based planning with **ReAct (Reasoning-Action)** framework, improving debugging accuracy by **45%** and reducing engineer troubleshooting time significantly.

Awards

Gemini Hack Night *Nov 2025*
- Secured 1st place among **150+ participants** by deploying an **AR based CPR assistance** web app in 5 hours using **Google MediaPipe**, **Gemini Pro**, and **ReactJS**, achieving **92%** measurement accuracy and **180ms** inference time.

Publications

- *"A Comparative Analysis of Transformer Models for Abstractive Summarization of Indian Legal Judgments"*, 1st International Conference on Machine Intelligence and Data Science (MIDAS 2025), Paralakhemundi, Odisha, India. Springer.